

wire' configuration allowed them to use membrane pores as the only pathway for the ions to reach the nanowire. This is how they could use the nanowire device to monitor specific transport and also to control the membrane protein.

The team showed that by changing the gate voltage of the device, they could open and close the membrane pore electronically.

## Circuit running off tree power

There is enough power in trees for University of Washington (UW) researchers to run an electronic circuit, according to the results to be published in an upcoming issue of the *Institute of Electrical and Electronics Engineers' Transactions on Nanotechnology*.

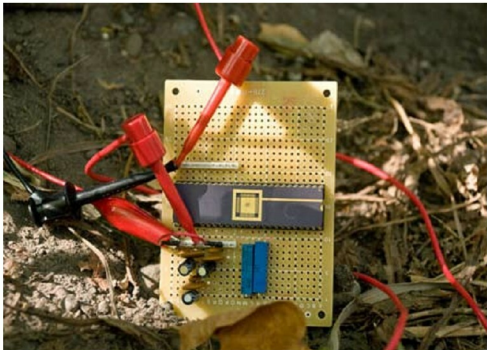
A study last year from the Massachusetts Institute of Technology had found that plants generate a voltage of up to 200 millivolts when one electrode is placed in a plant and the other in the surrounding soil. Those researchers have since started a company developing forest sensors that exploit this new power source.

The UW team sought to further research in the field of tree power by building circuits to run off that energy. Hooking nails to trees and connecting a voltmeter, they found that bigleaf maples—common on the UW campus—generate a steady voltage of up to a few hundred millivolts. The team next developed a boost converter—a device that takes a low incoming voltage and stores it to produce a greater output. The boost converter works off an input voltage as low as 20 millivolts—lower than for any such device—and outputs 1.1 volts, which is enough to run low-power sensors.

The UW circuit is built from parts measuring 130 nanometres and consumes on an average just 10 nanowatts of power during operation.

## Scientists photograph a single molecule

IBM scientists have been able to image the anatomy—or chemical structure—



The custom circuit developed by University of Washington researchers that can store up enough voltage from trees to run a low-power sensor

inside a molecule with unprecedented resolution. They achieved this feat by using a complex technique known as non-contact atomic force microscopy (AFM). The results push the exploration of using molecules and atoms at the smallest scale and could greatly impact the field of nanotechnology, which seeks to understand and control some of the smallest objects known to mankind.

The team's current achievement follows on the heels of another experiment where IBM scientists measured the charge states of atoms using AFM. These breakthroughs will open new possibilities for investigating how charge transmits through molecules or molecular networks. Understanding the charge distribution at the atomic scale is essential for building smaller, faster and more energy-efficient computing components than today's processors and memory devices.

## Computers of the future

Computers that use light to process large amounts of data faster than ever before could soon become possible, thanks to a research programme at Queen's University Belfast and Imperial College London.

The two universities are researching so-called 'nanoplasmonic devices.' The key components of nanoplasmonic devices are tiny nanoscale metal structures—more than 100 times smaller than the width of a human hair—that guide and direct light. The structures have been tailor-made to interact with light in an unusual and highly controlled way. This means they could be used to build new kinds of super-high-speed 'optical computers' that process information using light signals, instead of the electric currents used by today's computers.

At present, the speed with which computers process information is limited by the time it takes for the information to transfer between electronic components. Currently, this information is transferred using nanoscale metallic wires that transmit the signals as an electric current.

To speed up the process, the scientists hope to develop a way of sending the signals along the same wires in the form of light. In order to achieve this, they are developing a raft of new metallic devices including tiny nanoscale sources of light—nanoscale 'waveguides' to guide light along a desired route and nanoscale detectors to pick up the light signals.